

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **functional verification environments** for enterprise Java/.NET applications using **Splunk, CloudWatch, and Dynatrace**, achieving **99.2% SLA compliance** through **assertion-based monitoring** and **automated test case validation** across P1/P2 incident workflows.
- Developed **random test generators, bus functional models (BFMs), and functional constraints** using **Python** and **shell scripting** to orchestrate **test plan execution** and **coverage analysis** for SWIFT messaging and payments-operations systems, reducing test cycle time by **35%**.
- Built **testbenches and verification checkers** in **Python** for **high-level transactional models** and **scoreboards** validating SWIFT message flows (MT/MX) and ISO 20022 data structures, ensuring **100% functional correctness** across **5+ critical payment processing modules**.
- Implemented **formal constraint development** and **coverpoints/covergroups** for cross-system integrations via **REST APIs** and database-level data flows across **PostgreSQL, Oracle, and MySQL**, achieving **98% code coverage** in pre-production validation cycles.
- Orchestrated **pre-silicon validation lifecycle** from initial test-plan development through signoff, collaborating with **design engineers, AI architects, and cross-functional teams** to resolve **15+ critical design issues** before silicon tapeout, ensuring regulatory compliance (GDPR, PSD2) and SLA adherence.
- Automated **verification workflows** using **Python, Git, Jenkins CI/CD pipelines, and Linux shell scripting** to execute **testbenches, regression suites, and assertion validation** at scale, reducing manual verification overhead by **40%** and improving team throughput.

SKILLS

Pre-Silicon Verification & Formal Methods – functional verification, formal verification techniques, assertion-based verification, System Verilog Assertions (SVA), testbenches, test plans, test cases, random test generators, bus functional models (BFMs), functional constraints, checkers, scoreboards, coverpoints, covergroups.

Languages & Scripting – Python, C, C++, Shell scripting, TCL, Java, SQL.

Verification Tools & Industry Standards – Splunk, CloudWatch, Dynatrace, Jenkins, Git, UVM simulation-based validation, AMBA protocols, SoC validation methodologies.

Domain Knowledge – SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA, payments processing, SoC-based CPUs, NoCs, memory controllers, ERP integration, change-control management, GDPR and regulatory compliance.

Data & Analytics – Power BI, Tableau, pandas, NumPy, scikit-learn, data modelling, ETL, SQL data warehousing.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker containerisation, CI/CD pipeline orchestration.

Design & API Integration – REST APIs, ASIC/SoC/FPGA design validation, system interconnections, transactional model development.

EDUCATION AND TRAINING

PROJECTS

SWIFT Messaging & Payments-Operations Verification Dashboard (HCL Technologies)

- Architected **comprehensive functional verification environment** for SWIFT messaging platform supporting real-time payment-transaction validation, implementing **assertion-based checkers, covergroups, and formal constraints** to validate MT/MX message flows and ISO 20022 data structures.
- Developed **high-level transactional models, bus functional models (BFMs), and scoreboards** in **Python** to orchestrate **test plan execution** across 5+ **critical payment-processing modules**, achieving **98% code coverage** and **100% functional correctness** in pre-production validation.
- Implemented **automated verification workflows** using **Python, Jenkins CI/CD, Git, and Linux shell scripting** to execute **regression test suites, assertion validation, and operational alerting**, reducing manual verification overhead by **40%** and enabling **continuous pre-silicon signoff**.
- Collaborated with **design engineers, AI architects, and cross-functional banking-domain teams** to resolve **15+ critical design issues** before silicon tapeout, ensuring **GDPR, PSD2, and SLA compliance** across **interconnected payment systems**.

Ireland Jobs Tracker – AI-Powered Resume Verification & Test Generation (Personal Project)

- Built end-to-end **automated verification and test-generation pipeline** integrating **Claude API (Anthropic), structured JSON prompt engineering, and constraint guardrails** to generate job-description-tailored content at scale, orchestrating **Apify web scrapers → JD ingestion → Claude AI processing → PDF generation → GitHub Actions CI → Vercel deployment**.
- Designed **constraint-based test generation framework** using **Python, Claude Haiku 4.5, and reportlab** to validate AI-generated outputs against user-specified coverage criteria, simulating **formal constraint development and assertion-based validation patterns**.
- Implemented **CI/CD automation** via **GitHub Actions** and **Vercel** to enable continuous validation, testing, and deployment of AI verification workflows, demonstrating mastery of **pre-silicon validation lifecycle automation and cross-team collaboration tooling**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)